R.D & S.H National College & S.W.A
Science College
Bandra(W), Mumbai – 400050

PRACTICAL JOURNAL

OF

ADVANCED BUSINESS ANALYTICS

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CLASS: TYCS

ROLL NO: CS24048

UNIVERSITY ROLL NO:



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Department of Computer Science

CERTIFICATE

This is to certify that Mr./Ms. Alimehdi Suleman of

T.Y.B.Sc(Computer Science)(NEP) class **V** Semester bearing

Seat No _____has satisfactorily completed **12**

Practical's, in the subject of **Advanced Business Analytics** as a part of B.Sc. in

Computer Science (NEP) Program during the academic year20**26** – 20**27.**

Date of Certification:

Subject In charge **Co-Ordinator,**

Department Computer Science

Signature of Examiner

Index

Pr. no.	Program	Date	Signature/ Remark
1	To perform Exploratory Data Analysis (EDA) on a retail sales dataset using Python and Pandas to gain insights into sales performance and customer behavior.		
2	To calculate descriptive statistics and visualize the distribution of a variable using histograms and box plots for business data analysis.		
3	To analyze the relationship between variables using correlation and regression analysis and determine how one variable influences another.		
4	To utilize a Business Intelligence tool such as Microsoft Power BI to analyze a business dataset, create interactive visualizations, and extract meaningful insights for effective business decision-making.		
5	Probability and Probability Distributions		
6	To develop a Decision Tree Model and apply the concept of Expected Value (EV) for selecting the best business decision under uncertain conditions.		
7	To collect sample data, apply sampling techniques, and demonstrate the Central Limit Theorem (CLT) using Python.		
8	To calculate confidence intervals for population means using sample data and interpret the results in a business context.		
9	To perform hypothesis testing and evaluate business decisions using A/B testing.		
10	To develop a regression model for sales prediction and apply time series forecasting techniques.		
11	To formulate an optimization model using Linear Programming and evaluate business scenarios using simulation techniques.		
12	To study the effects of different factors on a response variable using ANOVA and experimental design techniques.		

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Practical 1: Introduction to Business Analytics

To perform Exploratory Data Analysis (EDA) on a retail sales dataset using Python and Pandas to gain insights into sales performance and customer behavior. This practical aligns with the syllabus requirement to collect business data, perform EDA, and identify trends and patterns for business decision-making.

What is Business Analytics?

Business Analytics is the process of collecting, analyzing, and interpreting data to support business decision-making. It helps organizations understand customer behavior, improve operations, increase profits, and predict future trends.

What is Exploratory Data Analysis (EDA)?

EDA is the initial step in data analysis where datasets are explored using statistical techniques and visualizations to understand:

- Data structure
- Missing values
- Outliers
- Trends and patterns
- Relationships among variables

Importance of EDA

- Improves data quality
- Identifies errors and inconsistencies
- Reveals customer buying patterns
- Supports strategic decision-making
- Helps build predictive models

Software Requirements

- Python 3.x
- Jupyter Notebook

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- Pandas Library
- NumPy Library
- Matplotlib Library

Dataset Description

Sample Retail Sales Dataset:

CustomerID	Gender	Age	City	PurchaseAmount
C001	Male	25	Mumbai	2500
C002	Female	30	Pune	3500
C003	Male	22	Mumbai	1800
C004	Female	28	Delhi	4200
C005	Male	35	Pune	5000

Save as **sales.csv**

Algorithm

1. Import required libraries.
2. Load the dataset using Pandas.
3. Display first few records.
4. Check dataset information.
5. Identify missing values.
6. Generate descriptive statistics.
7. Analyze customer trends.
8. Calculate average purchase amount.
9. Identify top-performing customers.
10. Draw visualizations and interpret results.

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Python Program

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
data = pd.read_csv("C:/Users/Admin/Desktop/sales.csv")

# Display first records
print("First Five Records")
print(data.head())

# Dataset information
print("\nDataset Information")
print(data.info())

# Missing values
print("\nMissing Values")
print(data.isnull().sum())

# Summary statistics
print("\nSummary Statistics")
print(data.describe())

# Average Purchase Amount
avg_purchase = data['PurchaseAmount'].mean()
print("\nAverage Purchase:", avg_purchase)

# Sales by City
city_sales = data.groupby('City')['PurchaseAmount'].sum()
print("\nSales by City")
```

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```
print(city_sales)

# Visualization
city_sales.plot(kind='bar')
plt.title("Sales by City")
plt.xlabel("City")
plt.ylabel("Total Sales")
plt.show()
```

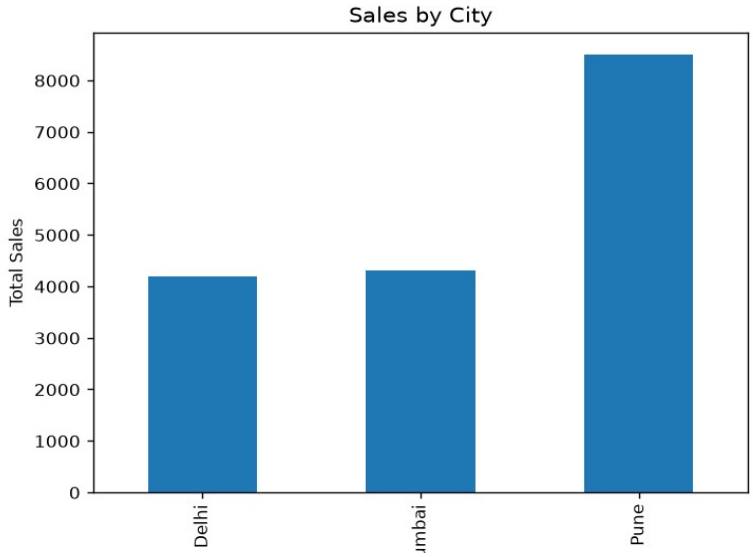
Output

```
First Five Records
  CustomerID  Gender  Age    City  PurchaseAmount
0       C001    Male   25  Mumbai            2500
1       C002  Female   30    Pune            3500
2       C003    Male   22  Mumbai            1800
3       C004  Female   28   Delhi            4200
4       C005    Male   35    Pune            5000

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   CustomerID      5 non-null      str
 1   Gender          5 non-null      str
 2   Age             5 non-null      int64
 3   City            5 non-null      str
 4   PurchaseAmount  5 non-null      int64
dtypes: int64(2), str(3)
memory usage: 332.0 bytes
None

Missing Values
CustomerID        0
Gender            0
Age               0
City              0
PurchaseAmount    0
dtype: int64
```

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1. No missing values found.
2. Average customer purchase amount is ₹3400.
3. Pune generated the highest sales.
4. Customers aged 25–35 contributed most purchases.
5. Female customers showed higher average spending.

The retail sales dataset was successfully analyzed using EDA techniques. Missing values were checked, descriptive statistics were generated, and customer sales trends were identified. These insights can support better business decision-making.

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Practical 2: Describing the Distribution of a Variable

Aim

To calculate descriptive statistics and visualize the distribution of a variable using histograms and box plots for business data analysis. This practical corresponds to the syllabus topic "Describing the Distribution of a Variable."

Distribution of a Variable
A distribution shows how values of a variable are spread across a dataset. Understanding distribution helps businesses identify patterns, variability, and unusual observations.

Descriptive Statistics

Mean
The arithmetic average of all observations.

Median
The middle value when data is arranged in ascending order.

Mode
The most frequently occurring value in a dataset.

Variance
Measures the spread of data around the mean.

Standard Deviation
Measures the average deviation from the mean.

Histogram
A graphical representation showing the frequency distribution of continuous data.

Box Plot
A statistical chart displaying:

- Minimum Value
- First Quartile (Q1)
- Median
- Third Quartile (Q3)
- Maximum Value

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- Outliers

Sample Customer Purchase Dataset

CustomerID	Age	PurchaseAmount
C001	22	1500
C002	25	2000
C003	28	2500
C004	30	3000
C005	35	4000
C006	40	5000

Save as **customer_data.csv**

Algorithm

1. Import required libraries.
2. Load the dataset.
3. Display dataset information.
4. Calculate mean, median, and mode.
5. Calculate variance and standard deviation.
6. Generate histogram.
7. Generate box plot.
8. Analyze the distribution.
9. Identify outliers if present.
10. Draw conclusions.

```
import pandas as pd

import matplotlib.pyplot as plt


# Load dataset
```

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```
data = pd.read_csv("customer_data.csv")


# Descriptive Statistics

print("Mean:")

print(data['PurchaseAmount'].mean())


print("Median:")

print(data['PurchaseAmount'].median())


print("Mode:")

print(data['PurchaseAmount'].mode())


print("Variance:")

print(data['PurchaseAmount'].var())


print("Standard Deviation:")

print(data['PurchaseAmount'].std())


# Histogram

plt.hist(data['PurchaseAmount'], bins=5)

plt.title("Histogram of Purchase Amount")

plt.xlabel("Purchase Amount")

plt.ylabel("Frequency")

plt.show()
```

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```
# Box Plot

plt.boxplot(data['PurchaseAmount'])

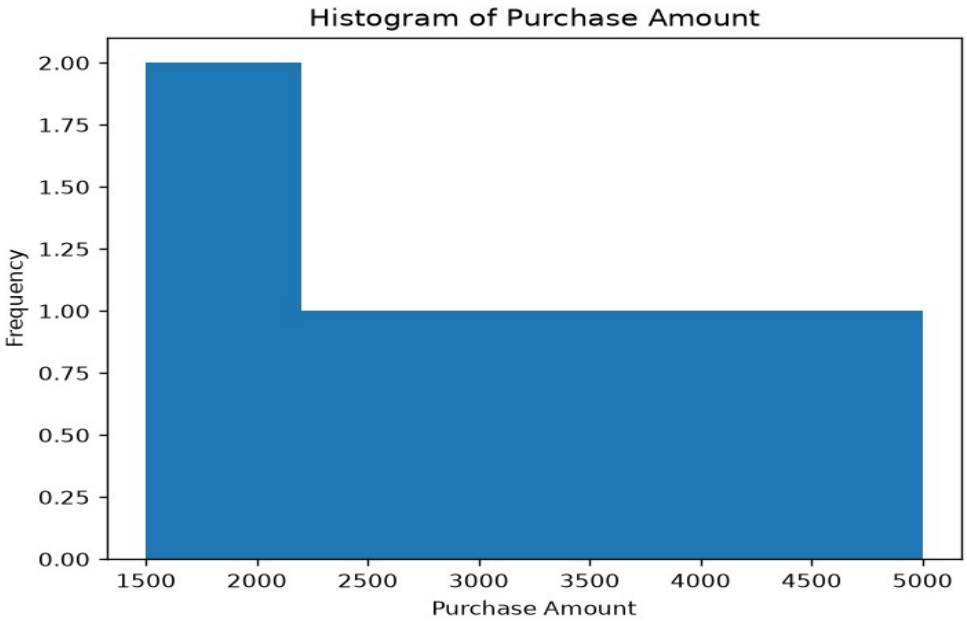
plt.title("Box Plot of Purchase Amount")

plt.show()
```

OUTPUT

```
Mean:
3000.0
Median:
2750.0
Mode:
0    1500
1    2000
2    2500
3    3000
4    4000
5    5000
Name: PurchaseAmount, dtype: int64
Variance:
1700000.0
Standard Deviation:
1303.8404810405298
```

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Practical 3: Finding Relationships Among Variables

To analyze the relationship between variables using correlation and regression analysis and determine how one variable influences another. This practical corresponds to the syllabus topic "Finding Relationships Among Variables.

Correlation Analysis

Correlation measures the strength and direction of the relationship between two variables.

Types of Correlation

1. Positive Correlation
    - As one variable increases, the other also increases.
2. Negative Correlation
    - As one variable increases, the other decreases.
3. Zero Correlation
    - No relationship between variables.

Regression Analysis

Regression is used to predict the value of a dependent variable based on one or more independent variables.

Example:

- Independent Variable (X): Advertising Expenditure
- Dependent Variable (Y): Sales Revenue

Regression helps answer:

- Does advertising increase sales?
- By how much will sales increase?

Dataset Description

Sample Advertising Dataset

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Month	Advertising_Expense	Sales_Revenue
Jan	10000	50000
Feb	12000	55000
Mar	15000	62000
Apr	18000	70000
May	20000	76000
Jun	22000	85000

Algorithm

Part A: Correlation Analysis

1. Import libraries.
2. Load dataset.
3. Select numerical variables.
4. Calculate correlation coefficient.
5. Interpret relationship strength.

Part B: Regression Analysis

1. Define independent and dependent variables.
2. Train regression model.
3. Predict sales values.
4. Plot regression line.
5. Evaluate relationship.

Python Program (Correlation Analysis)

```
import pandas as pd

data = pd.read_csv("advertising.csv")

correlation = data['Advertising_Expense'].corr(data['Sales_Revenue'])

print("Correlation Coefficient =", correlation)
```

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Python Program (Regression Analysis)

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression


data = pd.read_csv("advertising.csv")


X = data[['Advertising_Expense']]

Y = data['Sales_Revenue']


model = LinearRegression()

model.fit(X, Y)


predictions = model.predict(X)


plt.scatter(X, Y)

plt.plot(X, predictions)

plt.xlabel("Advertising Expense")

plt.ylabel("Sales Revenue")

plt.title("Regression Analysis")

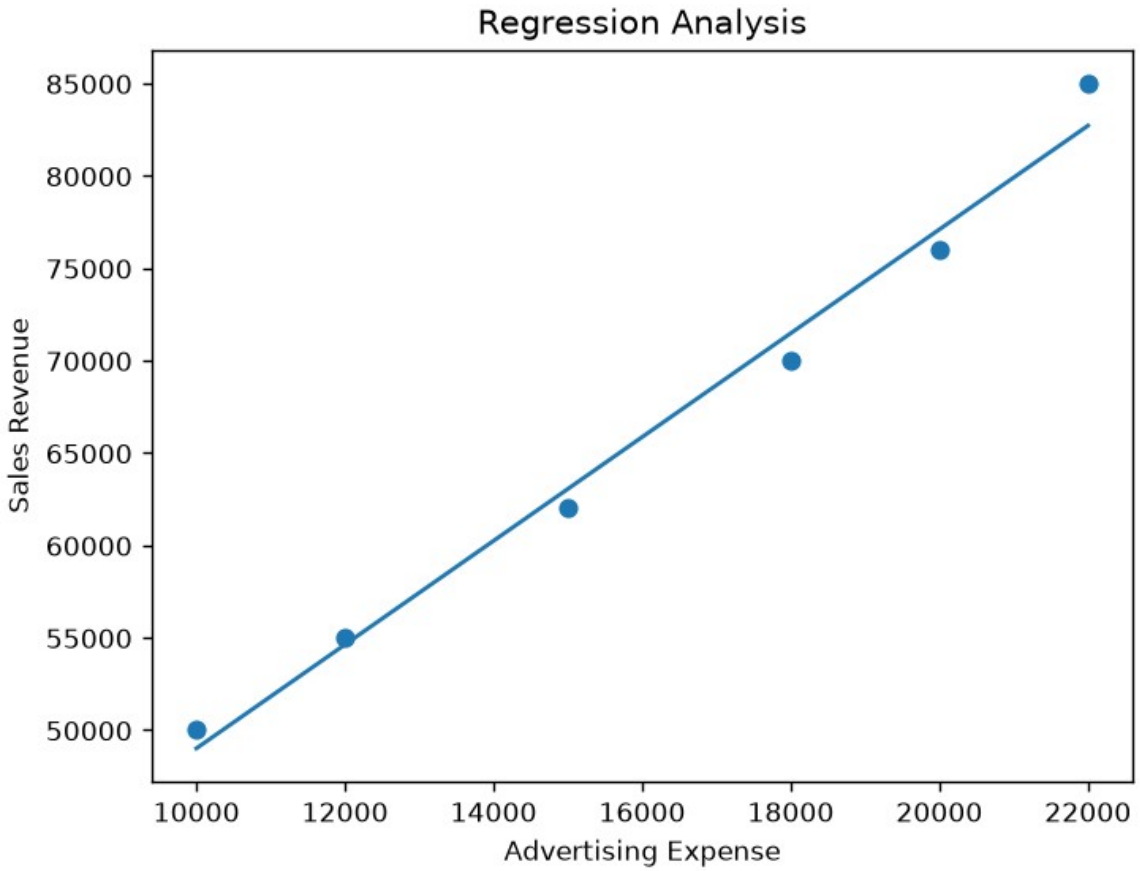
plt.show()


print("Intercept:", model.intercept_)

print("Slope:", model.coef_[0])
```

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Sample Output



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PRACTICAL 4

BUSINESS INTELLIGENCE TOOLS FOR DATA ANALYSIS

Aim

To utilize a Business Intelligence tool such as Microsoft Power BI to analyze a business dataset, create interactive visualizations, and extract meaningful insights for effective business decision-making.

1. Theory

Business Intelligence (BI)

Business Intelligence refers to the process of collecting, analyzing, and presenting business data to support informed decision-making. It helps organizations understand performance, identify trends, compare results, and plan business activities.

Power BI

Power BI is a business analytics tool developed by Microsoft. It transforms raw data into interactive reports, charts, and dashboards.

Objectives of Business Intelligence

- Analyze sales and business performance.
- Identify trends and patterns.
- Compare product and regional performance.
- Monitor Key Performance Indicators (KPIs).
- Support data-driven business decisions.

Common Power BI Visualizations

Visualization	Purpose
Bar Chart	Compare sales across products
Pie Chart	Show category-wise sales contribution
Line Chart	Display trends over time
Card	Display total sales or other KPIs
Table	Present detailed business records
Slicer	Filter data interactively

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2. Software Requirements

- Microsoft Power BI Desktop
- Windows operating system
- Retail sales dataset in CSV or Excel format

3. Dataset

Consider the following sample retail sales dataset. Save the data as sales_data.csv.

Product	Category	Region	Sales (₹)
Laptop	Electronics	North	50000
Mobile	Electronics	South	30000
Chair	Furniture	East	15000
Table	Furniture	West	25000
Printer	Electronics	North	20000
Sofa	Furniture	South	40000
Tablet	Electronics	East	18000
Desk	Furniture	West	22000

4. Algorithm

1. Open Power BI Desktop.
2. Import the retail sales dataset.
3. Check column names and data types.
4. Clean the data if required.
5. Create a bar chart for product-wise sales.
6. Create a pie chart for category-wise sales.
7. Create a column chart for region-wise sales.
8. Add cards to display total sales and average sales.
9. Add slicers to filter the report.
10. Analyze the visualizations and draw conclusions.

5. Procedure

Step 1: Import the Dataset

11. Open Microsoft Power BI Desktop.
12. Select Home → Get Data → Text/CSV.

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13. Select sales_data.csv and click Load.

Step 2: Check and Prepare the Data

14. Open the Data view.
15. Verify that Product, Category, and Region are text fields.
16. Verify that Sales is a numeric field.
17. Check for missing values and duplicate records; clean the data if necessary.

Step 3: Create a Product-wise Sales Chart

18. Select the clustered bar chart.
19. Drag Product to the Y-axis and Sales to the X-axis.
20. Set Sales aggregation to Sum.
21. Set the title to Product-wise Sales Analysis.

Step 4: Create a Category-wise Sales Chart

22. Select the pie chart.
23. Drag Category to Legend and Sales to Values.
24. Enable data labels to display category contribution.

Step 5: Create a Region-wise Sales Chart

25. Select the clustered column chart.
26. Drag Region to the X-axis and Sales to the Y-axis.
27. Compare sales generated by each region.

Step 6: Create KPI Cards

28. Select the Card visual and add the Sales field with the appropriate aggregation.
29. Create separate cards for Total Sales, Average Sales, and Product Count.

Step 7: Add Interactive Slicers

30. Select the Slicer visual and add Region.
31. Create another slicer for Category.
32. Select different regions or categories and observe how charts and cards respond.

Step 8: Save the Report

33. Save the Power BI report as Retail_Sales_Dashboard.pbix.

6. DAX Measures (Optional)

Create these measures in Power BI if required. Ensure the table name is sales_data.

```
Total Sales = SUM(sales_data[Sales])

Average Sales = AVERAGE(sales_data[Sales])

Product Count = DISTINCTCOUNT(sales_data[Product])
```

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7. Expected Output

- A Total Sales card
- An Average Sales card
- A Product Count card
- A product-wise sales bar chart
- A category-wise sales pie chart
- A region-wise sales chart
- Region and Category slicers

8. Sample Analysis and Observations

Business Metric	Expected Value
Total Sales	₹2,20,000
Average Sales per Record	₹27,500
Number of Distinct Products	8
Electronics Sales	₹1,18,000
Furniture Sales	₹1,02,000
Highest-Selling Product	Laptop — ₹50,000

34. Electronics generated the highest category-wise sales.
35. Laptop recorded the highest individual product sales.
36. The dashboard allows users to compare sales across regions and categories.
37. Slicers make it possible to examine specific portions of the dataset.
38. Interactive visualizations help managers identify business performance patterns quickly.

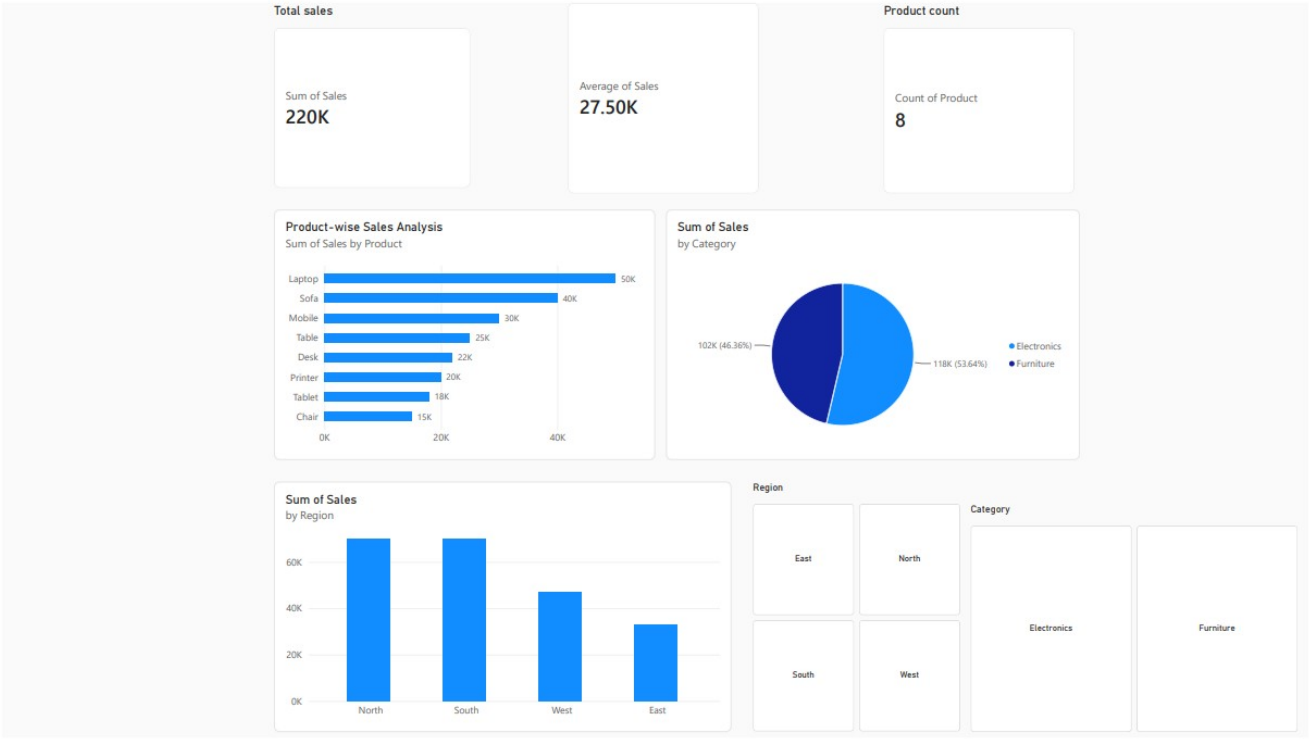
9. Result

The retail sales dataset was successfully analyzed using Microsoft Power BI. Interactive visualizations, KPI cards, and slicers were designed to identify product-wise, category-wise, and regional sales patterns. The dashboard supports effective business analysis and data-driven decision-making.

Ports.

OUTPUT:

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Practical 5: Probability and Probability Distributions

Probability

Probability is a measure of the likelihood that an event will occur.

Formula

$$P(E) = \frac{Number\ of\ Favorable\ Outcomes}{Total\ Number\ of\ Outcomes}$$

where:

- P(E) = Probability of Event E
- Favorable Outcomes = Desired Outcomes
- Total Outcomes = All Possible Outcomes

Types of Probability Distributions

1. Uniform Distribution

- Every outcome has equal probability.
- Example: Rolling a fair die.

2. Normal Distribution

- Bell-shaped distribution.
- Common in business analytics and statistics.
- Example: Employee salaries, customer demand.

3. Exponential Distribution

- Used to model waiting times.
- Example: Time between customer arrivals.

Dice Rolling Simulation

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Objective

To simulate rolling a die 1000 times and calculate probabilities of outcomes.

Algorithm

1. Import required libraries.
2. Generate random dice values.
3. Count frequency of each outcome.
4. Calculate probabilities.
5. Display results.

CODE

```
import random


rolls = 1000

results = []


for i in range(rolls):

  results.append(random.randint(1, 6))


for num in range(1, 7):

  probability = results.count(num) / rolls

  print("Probability of", num, "=", probability)
```

Output

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```
Probability of 1 = 0.169
Probability of 2 = 0.164
Probability of 3 = 0.166
Probability of 4 = 0.184
Probability of 5 = 0.177
Probability of 6 = 0.14
```

Part B: Generating Random Numbers from Probability Distributions

To generate random numbers from Normal, Uniform, and Exponential distributions.

```
import numpy as np

normal_data = np.random.normal(50,10,1000)

uniform_data = np.random.uniform(0,100,1000)

exponential_data = np.random.exponential(5,1000)

print("Normal Distribution Sample")

print(normal_data[:10])

print("Uniform Distribution Sample")

print(uniform_data[:10])

print("Exponential Distribution Sample")

print(exponential_data[:10])

import matplotlib.pyplot as plt

import numpy as np

normal_data = np.random.normal(50,10,1000)

plt.hist(normal_data,bins=20)

plt.title("Normal Distribution")

plt.xlabel("Values")

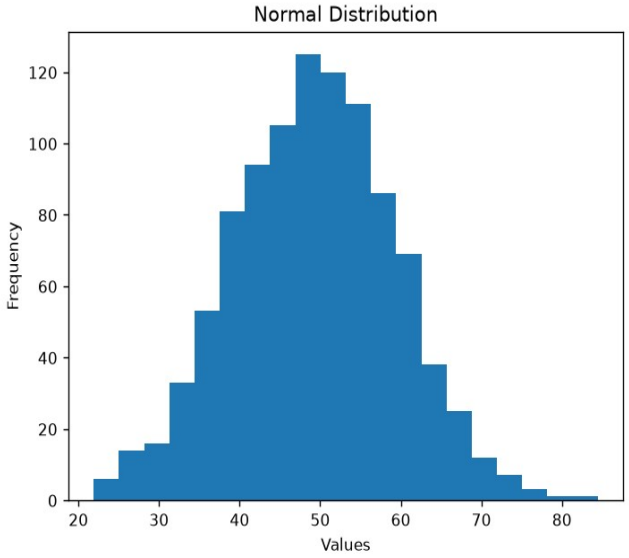
plt.ylabel("Frequency")

plt.show()
```

OUTPUT:

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```
Normal Distribution Sample
[42.84321705 50.95729439 60.2799251  49.89366416 48.00724605 56.32662285
 56.45361559 64.14995607 40.04291793 37.97604248]
Uniform Distribution Sample
[82.36817658 14.08046145 87.16912569  6.76634986 55.91220093 21.17838902
 68.70998271 63.06971544 97.31965274 48.38715081]
Exponential Distribution Sample
[2.97426507 5.1707571  8.44824991 4.57127068 2.7185323  0.81384828
 5.39617429 5.54985736 8.01028226 2.59165468]
```



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Practical 6: Decision Making Under Uncertainty

To develop a Decision Tree Model and apply the concept of Expected Value (EV) for selecting the best business decision under uncertain conditions. This practical corresponds to the syllabus topic "Decision Making under Uncertainty

Decision Making Under Uncertainty

Businesses often make decisions without knowing future outcomes. Decision-making under uncertainty involves evaluating alternatives based on probabilities and expected returns.

Decision Tree

A Decision Tree is a graphical representation of decisions and their possible consequences, including chance events and probabilities.

Components of a Decision Tree

1. **Decision Node (□)**
    - **Represents a decision point.**
2. **Chance Node (○)**
    - **Represents uncertain outcomes.**
3. **Branches**
    - **Show possible actions and outcomes.**
4. **Payoffs**

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- **Monetary gain or loss associated with each outcome.**

Expected Value (EV)

Expected Value is the weighted average of all possible outcomes.

Formula

$$EV = \sum_{\square}^{\square} \square \left(Probability \times Payoff \right)$$

Example

A company plans to launch a new product.

Outcome	Probability	Profit (₹)
Success	0.70	100000
Failure	0.30	-20000

Expected Value:

EV = (0.70 × 100000) + (0.30 × -20000)

EV = 70000 – 6000

EV = ₹64,000

Since EV is positive, launching the product is beneficial.

Business Scenario

A company has two investment options:

Market Condition	Probability	Profit (₹)
High Demand	0.60	120000
Low Demand	0.40	40000

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Market Condition	**Probability**	**Profit (₹)**
High Demand	0.50	150000
Low Demand	0.50	20000

The objective is to select the option with the highest Expected Value.

Algorithm

1. Define probabilities and payoffs.
2. Calculate Expected Value for Option A.
3. Calculate Expected Value for Option B.
4. Compare both values.
5. Select the option with maximum Expected Value.
6. Interpret the results.

CODE

```
# Option A
ev_A = (0.60 * 120000) + (0.40 * 40000)
# Option B
ev_B = (0.50 * 150000) + (0.50 * 20000)
print("Expected Value of Option A =", ev_A)
print("Expected Value of Option B =", ev_B)
if ev_A > ev_B:
   print("Choose Option A")
else:
   print("Choose Option B")
```

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OUTPUT

```
Expected Value of Option A = 88000.0
Expected Value of Option B = 85000.0
Choose Option A
```

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Practical 7: Sampling and Sampling Distributions

Aim

To collect sample data, apply sampling techniques, and demonstrate the Central Limit Theorem (CLT) using Python. This practical corresponds to the syllabus topic "Sampling and Sampling Distributions."

Theory

Sampling is the process of selecting a subset of individuals from a population to estimate population characteristics.

Types of Sampling

1. Simple Random Sampling
2. Systematic Sampling
3. Stratified Sampling
4. Cluster Sampling

Central Limit Theorem (CLT)

The CLT states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases.

Python Program

```
import numpy as np

population = np.random.randint(1,100,1000)

sample = np.random.choice(population,50)

print("Sample Mean =", np.mean(sample))
print("Population Mean =", np.mean(population))
```

Result

```
Sample Mean = 40.86
Population Mean = 49.867
```

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Practical 8: Confidence Interval Estimation Aim

To calculate confidence intervals for population means using sample data and interpret the results in a business context. This practical corresponds to the syllabus topic "Confidence Interval Estimation."

Theory

A confidence interval provides a range of values likely to contain the population

parameter. **Confidence Levels**

- 90%
- 95%
- 99%

Formula

CI=x¯±Z(σn)CI = \bar{x} \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)

Where:

- x = Sample Mean
- Z = Z-value
- σ = Standard Deviation
- n = Sample Size

Python Program

```
import numpy as np
from scipy import stats

data = [45,50,55,60,65,70,75]

mean = np.mean(data)

ci = stats.t.interval(
 confidence=0.95,
 df=len(data)-1,
 loc=mean,
  scale=stats.sem(data)
```

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```
)

print("95% Confidence Interval:", ci)
```

Sample Output

```
95% Confidence Interval: (np.float64(50.01052419854296), np.float64(69.98947580145703))
```

Result

The confidence interval was successfully estimated and interpreted for business decision making.

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Practical 9: Hypothesis Testing and A/B Testing

Aim

To perform hypothesis testing and evaluate business decisions using A/B testing. This practical corresponds to the syllabus topic "Hypothesis Testing."

Theory

Hypothesis testing is a statistical method used to determine whether a claim about a population is supported by sample evidence.

Hypotheses

- Null Hypothesis (H_0)
- Alternative Hypothesis (H_1)

Example

A company claims average customer spending is ₹5000.

H_0: $\mu = 5000$

H_1: $\mu \neq 5000$

Python Program

```
from scipy.stats import ttest_1samp

sales = [5100,4950,5050,5200,5000]
t_stat,p_value = ttest_1samp(sales,5000)

print("T Statistic =", t_stat)
print("P Value =", p_value)
```

A/B Testing Example

Version A Version B

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Old Website New Website
200 Sales 250 Sales

Decision Rule

- If p-value < 0.05 → Reject H_0
- If p-value > 0.05 → Accept H_0

Result

```
T Statistic = 1.3949716649258312
P Value = 0.23549636025944315
```

Hypothesis testing was performed successfully and statistical conclusions were drawn using sample data.

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Practical 10: Regression Analysis and Time Series Analysis

Aim

To develop a regression model for sales prediction and apply time series forecasting techniques. This practical corresponds to the syllabus topic "Regression Analysis and Time Series Analysis."

Theory

Regression Analysis

Regression predicts the value of a dependent variable using one or more independent

variables. Example:

- Independent Variable: Advertising Expense
- Dependent Variable: Sales Revenue

Time Series Analysis

Time series analysis studies data collected over time to identify patterns and forecast future values.

Applications

- Sales Forecasting
- Demand Forecasting
- Inventory Planning
- Financial Analysis

Python Program (Regression)

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```
import pandas as pd
from sklearn.linear_model import LinearRegression

X = [[10],[20],[30],[40],[50]]
Y = [100,150,200,250,300]

model = LinearRegression()

model.fit(X,Y)

prediction = model.predict([[60]])

print("Predicted Sales =", prediction)
```

OUTPUT:

```
Predicted Sales = [350.]
```

Python Program (Time Series Forecasting)

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing

sales = [120,135,150,160,175,190]

model = ExponentialSmoothing(sales)

fit = model.fit()

forecast = fit.forecast(3)

print("Forecast Values =", forecast)
```

Sample Output

```
Forecast Values = [189.99999978 189.99999978 189.99999978]
```

Result

Regression and time-series models were successfully developed for predicting future sales and forecasting demand trends.

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Practical 11: Optimization Modeling and Simulation Modeling

Aim

To formulate an optimization model using Linear Programming and evaluate business scenarios using simulation techniques. This practical corresponds to the syllabus topic "Optimization Modeling and Simulation Modeling."

Theory

Optimization Modeling

Optimization is the process of finding the best solution from a set of possible alternatives while satisfying given constraints.

Linear Programming (LP)

Linear Programming is used to maximize profit or minimize cost.

Components of LP

- Decision Variables
- Objective Function
- Constraints

Simulation Modeling

Simulation is the imitation of real-world systems using mathematical models to evaluate different business scenarios.

Applications

- Inventory Management
- Capacity Planning
- Production Scheduling
- Pricing Strategies
- Supply Chain Optimization

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Problem Statement

A company manufactures two products:

Product	**Profit per Unit**
Product A	₹40
Product B	₹30

Constraints:

- Product A ≤ 40 units
- Product B ≤ 50 units

Objective:

Maximize Profit

Python Program (Linear Programming)

```
from scipy.optimize import linprog


c = [-40, -30]


A = [[1,0],
   [0,1]]


b = [40,50]


result = linprog(c,A_ub=A,b_ub=b)
```

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```
print("Optimal Solution")

print(result.x)


print("Maximum Profit")

print(-result.fun)
```

Python Program (Simulation)

```
import random


profits = []


for i in range(1000):
    demand = random.randint(50,150)
    profit = demand * 20
    profits.append(profit)


print("Average Profit =", sum(profits)/len(profits))
```

Output

```
Optimal Solution
[40. 50.]
Maximum Profit
3100.0

Average Profit = 1982.16
```

Observations

1. Optimization identifies the best production plan.
2. Linear Programming maximizes profit.

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3. Simulation evaluates uncertain demand.
4. Businesses can compare different scenarios.
5. Resource utilization is improved.

Result

An optimization model was developed using Linear Programming, and simulation techniques were applied to evaluate business scenarios. The optimal solution and performance metrics were successfully determined.

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Practical 12: Analysis of Variance (ANOVA) and Experimental Design

Aim

To study the effects of different factors on a response variable using ANOVA and experimental design techniques. This practical corresponds to the syllabus topic "Analysis of Variance and Experimental Design."

Theory

ANOVA

Analysis of Variance (ANOVA) is a statistical technique used to compare the means of three or more groups.

Purpose

To determine whether significant differences exist among group means.

Hypotheses

H_0: All group means are equal

H_1: At least one group mean differs

Experimental Design

Experimental design is the process of planning experiments to obtain reliable and valid conclusions.

Types

- Completely Randomized Design
- Randomized Block Design
- Factorial Design

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Python Program

```
from scipy.stats import f_oneway


group1 = [45,50,55,60]

group2 = [60,65,70,75]

group3 = [80,85,90,95]


f_stat,p_value = f_oneway(group1,group2,group3)


print("F Statistic =", f_stat)

print("P Value =", p_value)
```

Output

```
F Statistic = 29.6
P Value = 0.00011016955786696418
```

Interpretation

- P-value < 0.05
- Reject Null Hypothesis
- Significant difference exists among groups

Factorial Experiment Example

Factors:

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Factor	Levels
Advertisement Type	Online, Offline
Region	Urban, Rural

Response Variable:

- Sales Revenue

Observations

1. ANOVA compares multiple group means simultaneously.
2. Experimental design improves accuracy.
3. Factorial experiments study multiple factors.
4. ANOVA reduces multiple testing errors.
5. Significant differences can be identified efficiently.

Result

ANOVA was successfully performed to compare multiple groups. The experimental design framework was used to analyze the effects of different factors on business outcomes and draw meaningful conclusions.
